

Main Vocabulary on Causal Inference and ML - Heterogeneous Treatment Effects

Kristian Boroz

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Heterogeneity and CATE

- **Conditional Average Treatment Effect (CATE):**

$$\tau(x) = \mathbb{E}[Y^{(1)} - Y^{(0)} \mid X = x].$$

The expected treatment effect for a subpopulation with specific characteristics X . *Goal: Move beyond population averages to granular effects.*

- **Treatment Heterogeneity:** the phenomenon where the causal impact of an intervention varies across different individuals or groups. *Example: A job training program helps young workers but has no effect on older workers.*
- **Subgroup Discovery:** searching for partitions of the data where treatment effects are significantly different. *Causal trees automate this to avoid "cherry-picking" results.*
- **Unconfoundedness:** the assumption that treatment assignment is independent of potential outcomes, conditional on observed covariates. *The "as good as random" requirement for observational data.*

Causal Trees and Forests

- **Causal Tree:** a recursive partitioning method that divides the covariate space into "leaves" to maximize the difference in treatment effects between groups. *Unlike regression trees, they optimize for treatment effect variance, not outcome variance.*
- **Honest Estimation:** a technique where the data is split into two parts: one to determine the tree structure (splits) and one to estimate the treatment effects within the leaves. *Prevents overfitting and ensures valid inference.*
- **Causal Forest:** an ensemble of many honest causal trees. It estimates $\tau(x)$ by averaging the predictions of individual trees. *Reduces the variance and "step-like" nature of a single tree.*
- **Adaptive Nearest-Neighbor:** the view that forests act as a weighting mechanism, giving more "weight" to observations that are similar in terms of relevant covariates. *A forest is like a k -NN that learns which dimensions of X actually matter.*
- **Curse of Dimensionality:** the problem where, in high-dimensional spaces, "nearby" points are actually very far apart. *Forests solve this by ignoring noise covariates and focusing on drivers of heterogeneity.*

Advanced Estimation and Diagnostics

- **Local Linear Forest:** a variation of the causal forest that applies a local linear correction to the leaf estimates. *Helps capture smooth, continuous trends rather than jagged step functions.*
- **Calibration:** a diagnostic process to check if the predicted $\hat{\tau}(x)$ matches the observed reality. *A "Best Linear Predictor" slope of 1 indicates a well-calibrated model.*
- **Calibration Plot:** a visualization (often using deciles) comparing predicted effects against actual effects on held-out data. *The 45-degree line represents perfect prediction.*
- **Asymptotic Normality:** the property that as the sample size grows, the distribution of the forest estimates approaches a normal distribution. *This allows for the construction of frequentist confidence intervals.*
- **False Discovery:** the risk of identifying treatment heterogeneity that is actually just statistical noise. *Honesty and cross-validation are the primary defenses against this.*

Common Shortnames

Shortname	Meaning
CATE	Conditional Average Treatment Effect
HTE	Heterogeneous Treatment Effects
MSE	Mean Squared Error
k-NN	k-Nearest Neighbors
GSS	General Social Survey (often used in experimental benchmarks)
BLP	Best Linear Predictor (used for calibration checks)